Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

J Am Coll Radiol. Author manuscript; available in PMC 2025 June 01.

Published in final edited form as:

J Am Coll Radiol. 2024 June ; 21(6): 858–866. doi:10.1016/j.jacr.2023.09.015.

Increased Utilization of Low-Dose Computed Tomography for Lung Cancer Screening at an Arkansas Community Oncology Clinic

Edgar T. Ellis, MPH[1], **Michael A. Bauer, PhD**[2], **J. Thaddeus Beck, MD**[3], **Daniel S. Bradford, MD**[3], **Joanna Thompson**[3], **Abby Holt, MPH, CTR**[4], **Margarete C. Kulik, PhD**[5,6], **Shelbie D. Stahr, PhD, MPH**[7], **Ping-Ching Hsu, PhD**[7], **L. Joseph Su, PhD, MPH**[8,#]

[1]Department of Epidemiology, Fay W. Boozman College of Public Health, University of Arkansas for Medical Sciences, Little Rock, AR 72205, USA;

[2]Department of Biomedical Informatics, College of Medicine, University of Arkansas for Medical Sciences, Little Rock, AR 72205, USA;

[3]Highlands Oncology, Fayetteville, AR 72703, USA;

[4]ICF International Inc., Fairfax, VA 22031, USA;

[5]Department of Health Behavior and Health Education, Fay W. Boozman College of Public Health, University of Arkansas for Medical Sciences, Little Rock, AR 72205, USA;

[6]Tobacco-Related Disease Research Program, University of California Office of the President, Oakland, California 94607, USA;

[7]Department of Environmental Health Sciences, Fay W. Boozman College of Public Health, University of Arkansas for Medical Sciences, Little Rock, AR 72205, USA;

[8]Peter O'Donnell Jr. School of Public Health, UT Southwestern Medical Center, Dallas, Texas 75390, USA

Abstract

[#]**Corresponding authors:** L. Joseph Su, Peter O'Donnell Jr. School of Public Health, UT Southwestern Medical Center, 5323 Harry Hines Blvd, Florence Building (E5.506X); Mail Code - 9066, Dallas, TX 75390-8876, Lihchyun.Su@UTSouthwestern.edu, Office: 214-648-6489.

Author contributions: **ETE**: Formal analysis, investigation, methodology, visualization, writing – original draft, writing – review & editing. **MAB**: Data curation, methodology. **JTB**: Conceptualization, project administration, resources. **DSB**: Conceptualization, project administration, resources. **JT**: Data curation, resources. **AH**: Data curation, formal analysis, methodology, visualization. **MCK**: Formal analysis, investigation, methodology, writing – original draft, review. **SDS**: Formal analysis, investigation, validation, visualization. **PCH**: Funding acquisition, resources, supervision, visualization, writing – review & editing. **LJS**: Conceptualization, data curation, formal analysis, investigation, methodology, project administration, resources, supervision, validation, visualization, writing – review & editing.

Publisher's Disclaimer: This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Data Statement

The author(s) declare(s) that they had full access to all of the data in this study and the author(s) take(s) complete responsibility for the integrity of the data and the accuracy of the data analysis.

Conflict of interest

The authors declare no conflict of interest.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Background: Low-dose computed tomography (LDCT) is underutilized in Arkansas for lung cancer screening, a rural state with high incidence of lung cancer. The objective was to determine whether offering free LDCT increased the number of high-risk individuals screened in a rural catchment area.

Methods: 5,402 patients enrolled in screening at Highlands Oncology, a community oncology clinic in Northwest Arkansas, from 2013 – 2020. Screenings were separated into time periods: Period 1 (10 months for-fee), Period 2 (10 months free with targeted advertisements and primary care outreach), and Period 3 (62 months free with only primary care outreach). 5,035 high-risk participants were eligible for analysis based on NCCN Clinical Practice Guidelines in Oncology. Enrollment rates, incidence densities (ID), Cox proportional hazard models, and Kaplan-Meier curves were performed to investigate differences between enrollment periods and high-risk groups.

Results: Patient volume increased drastically once screenings were offered free of charge (Period 1 = 4.6 vs. Period 2 = 66.0; Period 3 = 69.8 average patients per month). Incidence density per 1,000 person-years increased through each period ($ID_{Period\ 1}$ = 17.2; $ID_{Period\ 2}$ = 20.8; $ID_{Period\ 3}$ = 25.5 cases). Cox models revealed significant differences in lung cancer risk between high-risk groups (P = 0.012), but not enrollment periods (P = 0.19). Kaplan-Meier lung cancer-free probabilities differed significantly between high-risk groups (log-rank P = 0.00068), but not enrollment periods (log-rank P = 0.18).

Conclusions: This study suggests that eligible patients are more receptive to free LDCT screening, despite most insurances not having a required copay for eligible patients.

Summary Sentence

Patients in a rural catchment area with low access to care are more receptive to low-dose computed tomography screening for lung cancer when the services are offered free of charge, regardless of insurance status.

Keywords

Lung Cancer; Screening; LDCT; Rural

Introduction

The state of Arkansas (AR) has the second-highest incidence of lung cancer in the United States (US), with a largely rural and medically underserved population.[1] Ninety-one point eight men and 62.4 women per 100,000 people are diagnosed with lung cancer each year in Arkansas, with 2,950 new cases expected in 2023.[2] Low-dose computed tomography (LDCT) is a well-established procedure for lung cancer screening in large academic medical centers, although few local community oncology centers utilize this technology.

Early routine screening of high-risk patients with LDCT identifies more early-stage diseases and improves survival rates as much as fivefold.[3–6] In Arkansas, 25% of all lung cancer cases are diagnosed at an early stage, similar to 24% of all lung cancer cases diagnosed at an early stage nationally.[1] Participation in screening by eligible high-risk patients is still relatively low. In Arkansas, only 4% of those classified as high-risk were screened,

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

compared to the national average of 6%.[1] An evaluation of the 2017 Behavioral Risk Factor Surveillance System data based on 4,373 eligible patients found differences in participation rates of LDCT screening ranging from 6.5–18.1% among the ten states examined.[7] Barriers toward LDCT screening include lack of awareness, cost concerns, insurance coverage, patient perception, and access to screening.[8,9]

Many rural communities in Arkansas are medically underserved and often have a financial barrier to receive routine LDCT screening. Over 459,000 Arkansans (15.2%) lived below the federal poverty level in 2020, with rural counties having a higher poverty rate (17.6%) than urban counties (13.9%).[10] In Arkansas, Medicare covers lung cancer screening, whereas Medicaid does not.[11] Lack of insurance or patients who are underinsured are less likely to receive lung cancer screening[12] and may face significant financial burden for downstream imaging and treatment.[13] For Arkansans who have insurance, eligibility criteria set by the National Comprehensive Cancer Network (NCCN) Clinical Practice Guidelines in Oncology must be satisfied to qualify for screening. Once qualified, providers classify patients as High-Risk Group 1 or High-Risk Group 2.[14] The US Preventive Services Task Force (USPSTF) recommends annual lung cancer screening for specific high-risk populations, based on age and history of smoking.[15] Based on data from the National Health and Nutrition Examination Survey and National Health Interview Survey, as of 2010, 8.6 million people would fit the eligibility criteria recommended by the USPSTF.[16,17]

Highlands Oncology is a non-academic community cancer treatment center in Fayetteville, AR, with additional offices located in Northwest Arkansas (NWA). Highlands Oncology currently provides free LDCT lung cancer screening without requiring proof of insurance. Both unique aspects set it apart from academic medical centers that utilize LDCT. The present paper describes the screened patient population since the program's inception in May 2013 and compares patient enrollment volume, lung cancer detection rates, stages, and patient characteristics between the initial for-fee service and after switching to a free service.

Methods

Ethics and Legal Approval

The University of Arkansas for Medical Sciences Institutional Review Board determined that the secondary data analysis nature of this project is not human subject research as defined in 45 CFR 46.102, and therefore it does not fall under the jurisdiction of the IRB review process.(IRB#260988, April 15, 2020)

Study Population

Participants were eligible for screening if they met high-risk NCCN criteria determined by clinicians and lived in Highlands Oncology's catchment area encompassing four counties in NWA from May 2013 to February 2020. High-risk group 1 criteria require the participant to be 55 – 77 years old, having smoked 30 pack-years, being a current smoker or having quit less than 14 years ago.[15] High-risk group 2 criteria require the participant to be 50 years old or older, having smoked 20 or more pack-years, and having other risk factors.[15] Pack-years smoked were calculated by multiplying the number of packs of cigarettes a

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

participant smoked per day by the number of years smoked.[18,19] In 2021, the population size of the counties located in Highland's catchment area was as follows: Washington: 250,057; Benton: 293,692; Madison: 16,960; Carroll: 28,435.[20]

Participants were separated into three periods based on time of enrollment: Period 1 enrollment during the for-fee period, during which the program operated for a fee which included a copay depending on insurance coverage, or from $32 - $200 out-of-pocket and varied depending on whether the patient received one or two scans in a single visit (May 2013 – Feb 2014). Period 2 enrollment during the first ten months of free screening, independent of a qualifying patient's health insurance status (Mar 2014 – Dec 2014). Direct outreach programs were utilized to reach the at-risk population in NWA during Period 1 and 2. These outreach efforts included billboards, radio ads/interviews, direct mailings, newspaper and TV advertisements, and events with primary care physicians and their staff to promote the LDCT program. Period 3 enrollment during the five years of free screening with limited outreach, in which efforts switched to only PCP office outreach and TV advertisements during lung cancer awareness month (Jan 2015 – Feb 2020).

Highlands Oncology Screening Methods and Data Collection

The screening process was comprised of an initial risk assessment, followed by a first LDCT scan and further screenings at 3 or 12-month intervals. LDCT screenings were performed at Highlands Oncology during business hours (8 am – 5 pm), with half days on Saturdays depending on patient volume. Results were interpreted the same day by Highlands staff, and result letters were given to patients within five business days. Patients who were underinsured or uninsured were managed the same as those with insurance, and Highlands worked to help patients obtain insurance if needed for further follow-ups or treatment. Information about LDCT screening was given to patients on interpretation, and results were processed so that patients and referring physicians would understand screening results. If abnormal results were found, personalized recommendations from Highlands Oncology providers for future treatment would ensue.

Highlands Oncology staff collected demographic information from consenting patients at their initial visit to determine NCCN risk group classification. Demographic information included sex, age, smoking status, and total years smoked. Scans were performed by radiologists and cancer diagnoses were confirmed by oncologists at Highlands Oncology. Data upkeep throughout the program was conducted by trained Highlands Oncology professionals. Data analysis and manuscript preparation were performed by independent researchers at the University of Arkansas for Medical Sciences with permission of Highlands Oncology.

Statistical Analysis

Participant characteristics between enrollment periods were evaluated using chi-square tests for homogeneity for categorical variables which include gender, risk group, and smoking history. ANOVA F-tests were used for continuous variables which include participant age at completion of risk assessment and log-transformed pack-years smoked. Frequency procedures were used to determine the numbers and percentages of individuals diagnosed

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

with Stage I, Stage II-IV, and other cancers in the three recruitment periods. Incidence density for any cancer diagnosed was calculated for each period per 1,000 person-years. Other cancers include mediastinal or breast cancers, which are identifiable through LDCT scans.[21] Multivariable Cox proportional hazard models were performed for Stage I-IV lung cancer diagnoses, and cause-specific hazard models were performed for Stage I or Stage II-IV lung cancer diagnoses. Cox models were adjusted for age at completion of risk assessment, pack-years smoked, smoking status, NCCN risk group, and sex. Time was calculated as days in the screening program from completion of risk assessment to an event (lung cancer diagnosis) or censoring (death, patient leaving the program, or being diagnosed with a non-lung cancer). Period 3 was used as the reference for the Cox models. Kaplan-Meier curves were performed presenting days to lung cancer-free probability stratified by risk groups and by enrollment periods. A 2-tailed p-value < 0.05 was used to indicate statistical significance. All data analyses were performed using SAS statistical software package, version 9.4 (SAS Institute, Cary, NC). Kaplan-Meier curves and log-rank tests were performed using R statistical software (v4.3.0) via the survminer R package[22] (v0.4.9).

Results

Patient Population and Demographics

5,398 patients participated in LDCT screening from May 2013 to February 2020. 367 participants were excluded from data analysis due to falling outside of the NCCN high-risk group guidelines. Patients included in the analysis fell into one of two high-risk NCCN Clinical Practice Guidelines in Oncology-defined risk groups[12]: risk group 1 (n = 3,905; 77.6%); risk group 2 (n = 1,126; 22.4%) for a total of 5,031 patients.

Patient characteristics and volume are presented in Table 1 and Figure 1. 46 patients participated in the for-fee program during Period 1, while 660 patients enrolled in the free program during Period 2. When the program was free with limited outreach efforts by Highlands (Period 3), 4,325 patients were enrolled. The number of current smokers was higher in Period 1 (65.2%) than in Period 2 (50.5%) or Period 3 (55.9%) ($P = 0.01$). Those enrolled in Period 1 had lower pack-years smoked (Median = 40.0; IQR = 25.0) than those enrolled in Period 2 (Median = 46.0; IQR = 24.0) or Period 3 (Median = 51.0; IQR = 39.5) ($P < .001$). Finally, patients enrolled in Period 1 were younger (Mean = 61.7; SD = 6.4) than those in Period 2 (Mean = 64.2; SD = 6.9) or Period 3 (Mean = 63.5; SD = 7.0) ($P = 0.01$).

Lung Cancer Diagnoses During Screening Periods

The total cancer cases detected during the screening periods, average enrollment rates per month, and cancer incidence density per 1,000 person-years are presented in Table 2. Among those who were enrolled and assessed during Period 1 (n = 46), two Stage I and one Stage II-IV lung cancer cases were diagnosed, with 4.6 patients enrolled per month and an incidence density of 17.2 cases per 1,000 person-years. Among those enrolled during Period 2 (n = 660), 21 Stage I, 10 Stage II-IV lung cancers, and 8 other cancers were diagnosed, with 66.0 patients enrolled per month and an incidence density of 20.8 cases per 1,000 person-years. Finally, among those who began enrollment in Period 3 (n = 4,325), 50 Stage I, 33 Stage II-IV lung cancers were diagnosed, and 25 other cancers were diagnosed,

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

with 69.8 patients enrolled per month and an incidence density of 25.5 cases per 1,000 person-years.

The unadjusted overall and cause-specific hazard ratios for Periods 1 and 2 were non-significant (Period 1 $HR_{Overall}$ = 1.72; 95% CI = 0.54 to 5.48; Period 2 $HR_{Overall}$ = 1.44; 95% CI = 0.95 to 2.20) (Period 1 $HR_{Stage\ I}$ = 1.83; 95% CI = 0.47 to 7.06; Period 2 $HR_{Stage\ I}$ = 1.59; 95% CI 0.93 to 2.69) (Period 1 $HR_{Stage\ II–IV}$ = 1.55; 95% CI = 0.23 to 10.62; Period 2 $HR_{Stage\ II–IV}$ = 1.21; 95% CI = 0.58 to 2.51). The adjusted overall and cause-specific hazard ratios are presented in Table 3 adjusted for age, log-transformed pack-years smoked, sex, smoking status, and risk group. Those in Periods 1 and 2 did not have a significantly different risk for lung cancer diagnosis than those in Period 3 (P = 0.19), but those in high-risk group 1 had a significantly higher risk for lung cancer diagnosis than those in high-risk group 2 (HR = 2.26; 95% CI 1.20 to 4.26; P = 0.01). Current smokers were also at higher risk for lung cancer than former smokers (HR = 1.69; 95% CI 1.14 to 2.51; P = 0.009). There was a 7.0% increase in the expected hazard of lung cancer relative to a one-year increase in age, holding all other variables constant (HR = 1.07; 95% CI 1.04 to 1.10; P < 0.0001). In the Stage I diagnosis cause-specific model, current smokers were at higher risk than former smokers for early-stage lung cancer (HR = 1.81; 95% CI 1.08 to 3.03; P = 0.02). There was a 6.0% increase in the expected hazard of Stage I lung cancer relative to a one-year increase in age, holding all other variables constant (HR = 1.06; 95% CI 1.02 to 1.10; P = 0.001). Finally, in the Stage II–IV diagnosis cause-specific model, high-risk group 1 individuals were at higher risk than high-risk group 2 individuals for late-stage lung cancer (HR = 4.06; 95% CI 1.33 to 12.35; P = 0.01). There was an 8.0% increase in the expected hazard of Stage II-IV lung cancer relative to a one-year increase in age, holding all other variables constant (HR = 1.08; 95% CI 1.02 to 1.14; P = 0.007).

Kaplan-Meier curves presenting lung cancer-free probability show that patients in risk group 1 had a significantly higher probability of being diagnosed with lung cancer than those in risk group 2 (log-rank P < .001) (Figure 2), while there was no significant difference between non-diagnosis probability across enrollment periods (log-rank P = 0.18) (Figure 3). Sensitivity analysis excluding Period 1 did not reveal significant differences between Period 2 and 3 (**Figure not shown**).

Discussion

Low-dose computed tomography screening programs have been demonstrated to reduce lung cancer deaths by up to 20% in high-risk populations.[23] However, the availability of LDCT facilities and uptake among high-risk persons is still low despite the 2013 recommendations by the USPSTF, the NCCN, and new reimbursement policies favoring LDCT screening.[24] This is particularly true among states with a majority rural population. This program evaluation of the implementation and practice of a non-academic community oncology clinic in Arkansas, a state with high lung cancer mortality but low access to LDCT screening,[25] demonstrated that eligible high-risk persons are more likely to participate in the screening program when the service is offered free of charge, regardless of insurance status. The number of patients increased from averaging 4.6 per month during the for-fee period to 66.0 per month after the clinic decided to offer the service free of charge. The number

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

of patients screened remained at an average of 69.8 patients per month even after the clinic stopped targeted advertisement to the at-risk population and focused mainly on primary care outreach and education. Although there is not an immediate shift in lung cancer diagnosis percentages from late-stage (stage II to IV) to early-stage, where cancer is more treatable, from Period 1 to Period 2 more patients were diagnosed with early-stage lung cancer (n = 22) than late-stage cancer (n = 10). Eight patients were diagnosed with other types of cancer in Period 2 versus none during Period 1.

The overall adjusted hazard model showed that being in risk group 1, being a current smoker, and older age were significantly associated with an increased risk of being diagnosed with any stage lung cancer. In the cause-specific models, being a current smoker and older age were significantly associated with increased risk for Stage I lung cancer, while being in risk group 1 and older age were significantly associated with increased risk for Stage II-IV lung cancer. Kaplan-Meier analysis showed that patients belonging to risk group 1 had a significantly higher probability of being diagnosed with lung cancer than those in risk group 2. Kaplan-Meier analysis by period of enrollment did not find any statistically significant difference in lung cancer-free probability between the periods, but with the more comparable Periods 2 and 3, Period 2 enrollees had a higher probability of being diagnosed with lung cancer than Period 3 enrollees.

There are examples of other community-based radiology clinics utilizing LDCT,[26,27] but most were part of screening trials. A 12-month free program in Pennsylvania identified specific challenges of lung cancer screening in a community setting in 2014/2015, with a total of 169 eligible participants for screening.[27] While the screening center served a diverse patient population, with about one-quarter of them living at or below 100% of the Federal Poverty Line, it is situated in Philadelphia, an urban setting with high population density. The challenges included a high non-compliance and no-show rate, insurance problems, and precertification. The Highlands Oncology screening program illustrates the success of a free LDCT screening program in a medically underserved state with low population density. It is unique compared to other clinics by offering LDCT screening entirely free and independent of a patient's health insurance status. This fact is significant as reimbursement for LDCT screenings follow strict guidelines with shared decision-making sessions, and insurance requirements may hinder initial screenings of high-risk individuals.[27,28]

Future studies are needed in Arkansas to evaluate the effects that participant awareness, income, access, and education have on LDCT screening rates. Future studies will be vital to investigate the possible unintended consequences of LDCT screening, including overdiagnosis, false positives, incidental findings, radiation exposure, and effects on overall quality of life.[16,29] Radiation exposure is increased for patients that are screened, but the harmful effects of radiation are minimal for older patients.[29] There are possible negative effects on quality of life when intermediate results are found.[30] Future studies are also needed due to the Centers for Medicare and Medicaid Services recently expanding their eligibility criteria for LDCT screening, changing the minimum age to 50, having smoked 20 pack-years, or being a current smoker or having quit smoking tobacco products in the past 15 years.[31]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

The goals of this project were to investigate whether offering free LDCT lung cancer screenings in a community oncology clinic increased patient volume and screenings of high-risk patients in a rural catchment area, and to determine the potential benefits of free versus for-fee screening. To continue increased screening rates in high-risk populations, community-based oncology centers providing free LDCT lung cancer screening with targeted advertisements and primary care outreach in otherwise medically underserved, rural areas are necessary.

Limitations

Limitations of this study include small sample size for Period 1 and uncontrolled potential confounders including race/ethnicity, individual/household income, insurance status at enrollment, other tobacco use, and occupation. Barriers to screening were not captured for this study, and future research may provide more insight into socioeconomic or demographic factors that affect screening rates in rural areas.[8,9] Assessment of risks and benefits of routine screening was beyond the scope of this study, and future studies would be beneficial for research into this topic. Potential biases include screening bias if those that chose to participate were more health conscious than those who did not. The results obtained from this study are generalizable to community oncology clinics in rural catchment areas, but rural/urban differences may affect generalizability to metropolitan areas that have larger population densities. However, these limitations do not affect the significance of results found for the aims and scope of this study.

Acknowledgements

The authors would like to thank the participants, whose contributions were vital for this study.

Funding

This project was supported by the COBRE Center for Studies of Host Response to Cancer Therapy, funded by the National Institute of General Medical Sciences (P20 GM109005 to PCH).

Role of the funders:

The funding sources had no role in the study design, the collection, analysis and interpretation of the data; the writing of the report; or the decision to submit the paper for publication.

Data Availability

The data underlying this article cannot be shared due to the privacy of individuals that participated in the study. Summary level data will be shared on request to the corresponding author with permission of Highlands Oncology.

References

1. State Data | Arkansas | American Lung Association. Accessed March 2, 2022. https://www.lung.org/research/state-of-lung-cancer/states/arkansas
2. Siegel RL, Miller KD, Wagle NS, Jemal A. Cancer statistics, 2023. CA Cancer J Clin. Jan 2023;73(1):17–48. doi:10.3322/caac.21763 [PubMed: 36633525]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

3. Henschke CI, McCauley DI, Yankelevitz DF, et al. Early Lung Cancer Action Project: overall design and findings from baseline screening. Lancet. Jul 10 1999;354(9173):99–105. doi:10.1016/S0140-6736(99)06093-6 [PubMed: 10408484]
4. Sone S, Takashima S, Li F, et al. Mass screening for lung cancer with mobile spiral computed tomography scanner. Lancet. Apr 25 1998;351(9111):1242–5. doi:10.1016/S0140-6736(97)08229-9 [PubMed: 9643744]
5. Swensen SJ, Jett JR, Hartman TE, et al. Lung cancer screening with CT: Mayo Clinic experience. Radiology. Mar 2003;226(3):756–61. doi:10.1148/radiol.2263020036 [PubMed: 12601181]
6. Flehinger BJ, Kimmel M, Melamed MR. The effect of surgical treatment on survival from early lung cancer. Implications for screening. Chest. Apr 1992;101(4):1013–8. doi:10.1378/chest.101.4.1013 [PubMed: 1313349]
7. Zahnd WE, Eberth JM. Lung Cancer Screening Utilization: A Behavioral Risk Factor Surveillance System Analysis. Am J Prev Med. Aug 2019;57(2):250–255. doi:10.1016/j.amepre.2019.03.015 [PubMed: 31248742]
8. Wang GX, Baggett TP, Pandharipande PV, et al. Barriers to Lung Cancer Screening Engagement from the Patient and Provider Perspective. Radiology. Feb 2019;290(2):278–287. doi:10.1148/radiol.2018180212 [PubMed: 30620258]
9. Simmons VN, Gray JE, Schabath MB, Wilson LE, Quinn GP. High-risk community and primary care providers knowledge about and barriers to low-dose computed topography lung cancer screening. Lung Cancer. Apr 2017;106:42–49. doi:10.1016/j.lungcan.2017.01.012 [PubMed: 28285693]
10. Agriculture ERSUSDo. State Fact Sheets: Arkansas. U.S. Department of Agriculture. Accessed March 2, 2022. https://data.ers.usda.gov/reports.aspx?StateFIPS=05&StateName=Arkansas&ID=17854
11. Pyenson BS, Henschke CI, Yankelevitz DF, Yip R, Dec E. Offering lung cancer screening to high-risk medicare beneficiaries saves lives and is cost-effective: an actuarial analysis. Am Health Drug Benefits. Aug 2014;7(5):272–82. [PubMed: 25237423]
12. Sun J, Perraillon MC, Myerson R. The Impact of Medicare Health Insurance Coverage on Lung Cancer Screening. Med Care. Jan 1 2022;60(1):29–36. doi:10.1097/MLR.0000000000001655 [PubMed: 34739415]
13. Slatore CG, Au DH, Gould MK, American Thoracic Society Disparities in Healthcare G. An official American Thoracic Society systematic review: insurance status and disparities in lung cancer practices and outcomes. Am J Respir Crit Care Med. Nov 1 2010;182(9):1195–205. doi:10.1164/rccm.2009-038ST [PubMed: 21041563]
14. Wood DE, Kazerooni EA, Baum SL, et al. Lung Cancer Screening, Version 3.2018, NCCN Clinical Practice Guidelines in Oncology. J Natl Compr Canc Netw. Apr 2018;16(4):412–441. doi:10.6004/jnccn.2018.0020 [PubMed: 29632061]
15. Moyer VA, Force USPST. Screening for lung cancer: U.S. Preventive Services Task Force recommendation statement. Ann Intern Med. Mar 04 2014;160(5):330–8. doi:10.7326/M13-2771 [PubMed: 24378917]
16. Jonas DE, Reuland DS, Reddy SM, et al. Screening for Lung Cancer With Low-Dose Computed Tomography: Updated Evidence Report and Systematic Review for the US Preventive Services Task Force. JAMA. Mar 9 2021;325(10):971–987. doi:10.1001/jama.2021.0377 [PubMed: 33687468]
17. Ma J, Ward EM, Smith R, Jemal A. Annual number of lung cancer deaths potentially avertable by screening in the United States. Cancer. Apr 1 2013;119(7):1381–5. doi:10.1002/cncr.27813 [PubMed: 23440730]
18. Definition of pack year - NCI Dictionary of Cancer Terms - NCI. Accessed October 10, 2022. https://www.cancer.gov/publications/dictionaries/cancer-terms/def/pack-year
19. Institute NC. NCI Dictionary of Cancer Terms. National Insitutes of Health. Accessed October 10, 2022. https://www.cancer.gov/publications/dictionaries/cancer-terms/def/pack-year
20. U.S. Census Bureau QuickFacts: Carroll County, Arkansas; Madison County, Arkansas; Benton County, Arkansas; Washington County, Arkansas; Arkansas. Accessed March 2, 2022. https://www.census.gov/quickfacts/fact/table/

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

carrollcountyarkansas,madisoncountyarkansas,bentoncountyarkansas,washingtoncountyarkansas,AR/PST045221

21. Yip R, Jirapatnakul A, Hu M, et al. Added benefits of early detection of other diseases on low-dose CT screening. Transl Lung Cancer Res. Feb 2021;10(2):1141–1153. doi:10.21037/tlcr-20-746 [PubMed: 33718052]
22. survminer: Drawing Survival Curves using 'ggplot2'. 2021. Accessed March 12, 2023. https://CRAN.R-project.org/package=survminer
23. National Lung Screening Trial Research T, Aberle DR, Adams AM, et al. Reduced lung-cancer mortality with low-dose computed tomographic screening. N Engl J Med. Aug 4 2011;365(5):395–409. doi:10.1056/NEJMoa1102873 [PubMed: 21714641]
24. Jemal A, Fedewa SA. Lung Cancer Screening With Low-Dose Computed Tomography in the United States-2010 to 2015. JAMA Oncol. Sep 1 2017;3(9):1278–1281. doi:10.1001/jamaoncol.2016.6416 [PubMed: 28152136]
25. Eberth JM, Bozorgi P, Lebron LM, et al. Geographic Availability of Low-Dose Computed Tomography for Lung Cancer Screening in the United States, 2017. Prev Chronic Dis. Oct 4 2018;15:E119. doi:10.5888/pcd15.180241 [PubMed: 30289107]
26. Byrne MM, Lillie SE, Studts JL. Lung cancer screening in a community setting: Characteristics, motivations, and attitudes of individuals being screened. Health Psychol Open. Jan-Jun 2019;6(1):2055102918819163. doi:10.1177/2055102918819163 [PubMed: 30643640]
27. Randhawa S, Drizin G, Kane T, Song GY, Reilly T, Jarrar D. Lung Cancer Screening in the Community Setting: Challenges for Adoption. Am Surg. Sep 1 2018;84(9):1415–1421. [PubMed: 30268168]
28. Aberle DR. Implementing lung cancer screening: the US experience. Clin Radiol. May 2017;72(5):401–406. doi:10.1016/j.crad.2016.12.003 [PubMed: 28069160]
29. Bach PB, Mirkin JN, Oliver TK, et al. Benefits and harms of CT screening for lung cancer: a systematic review. JAMA. Jun 13 2012;307(22):2418–29. doi:10.1001/jama.2012.5521 [PubMed: 22610500]
30. van den Bergh KA, Essink-Bot ML, Borsboom GJ, et al. Short-term health-related quality of life consequences in a lung cancer CT screening trial (NELSON). Br J Cancer. Jan 5 2010;102(1):27–34. doi:10.1038/sj.bjc.6605459 [PubMed: 19935789]
31. Screening for Lung Cancer with Low Dose Computed Tomography (LDCT) CAG-00439R (2022).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Take Home Points

- Low-dose computed tomography screening for lung cancer is underutilized in a rural catchment area of Arkansas with a low access to care population
- Offering low-dose computed tomography for lung cancer screening to high-risk patients for free with targeted advertising and primary care physician outreach, regardless of patient insurance status, led to a significant increase in screening volume compared to the traditional for-fee model in the catchment area
- High-risk patients in a rural catchment area are more receptive to low-dose computed tomography screening services offered free of charge, even when insurance would cover required copays under current Medicare guidelines

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

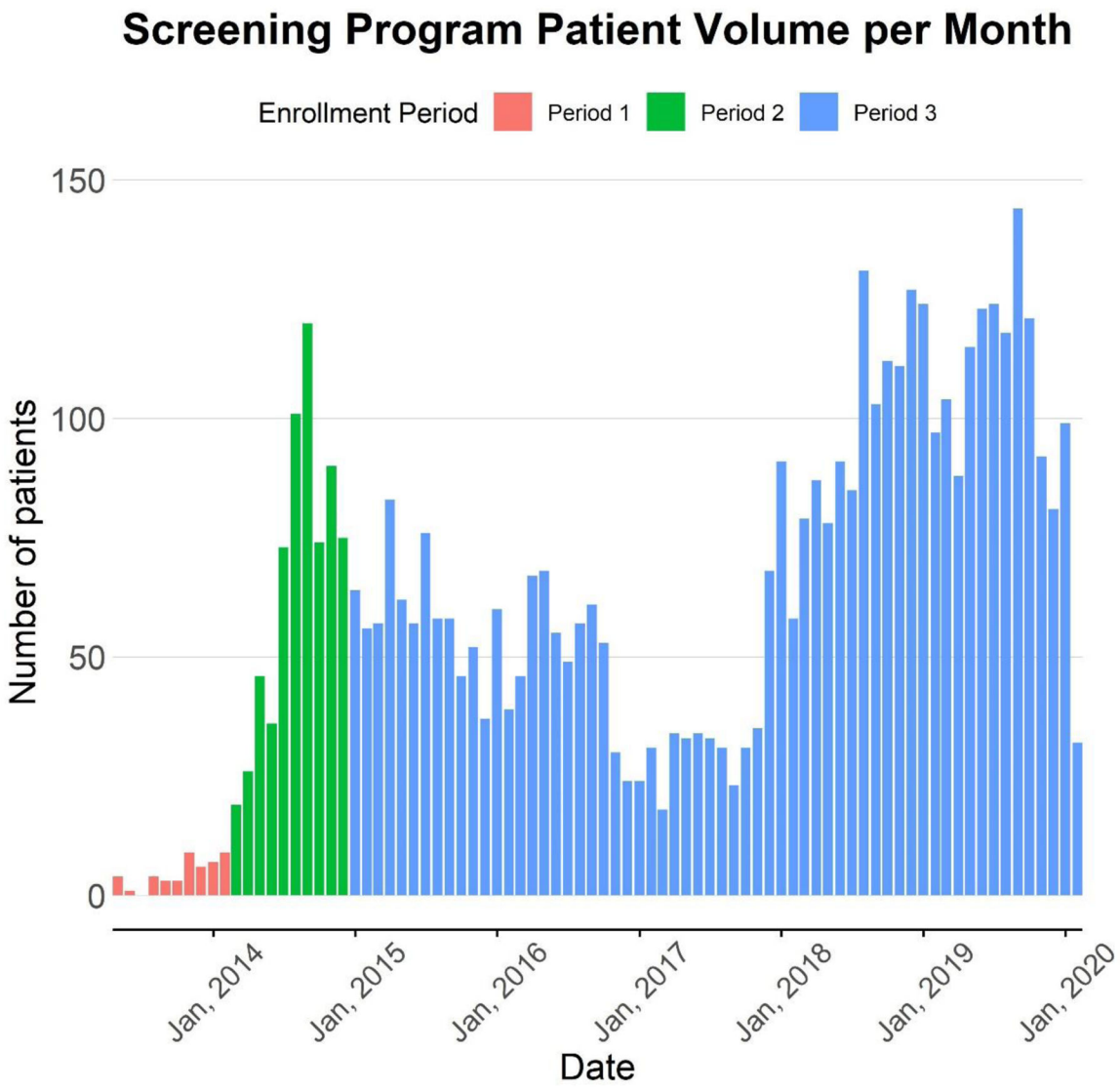


Figure 1.

Histogram of the Highlands Oncology screening program patient volume per month, colored by enrollment periods from May 2013 to February 2020.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

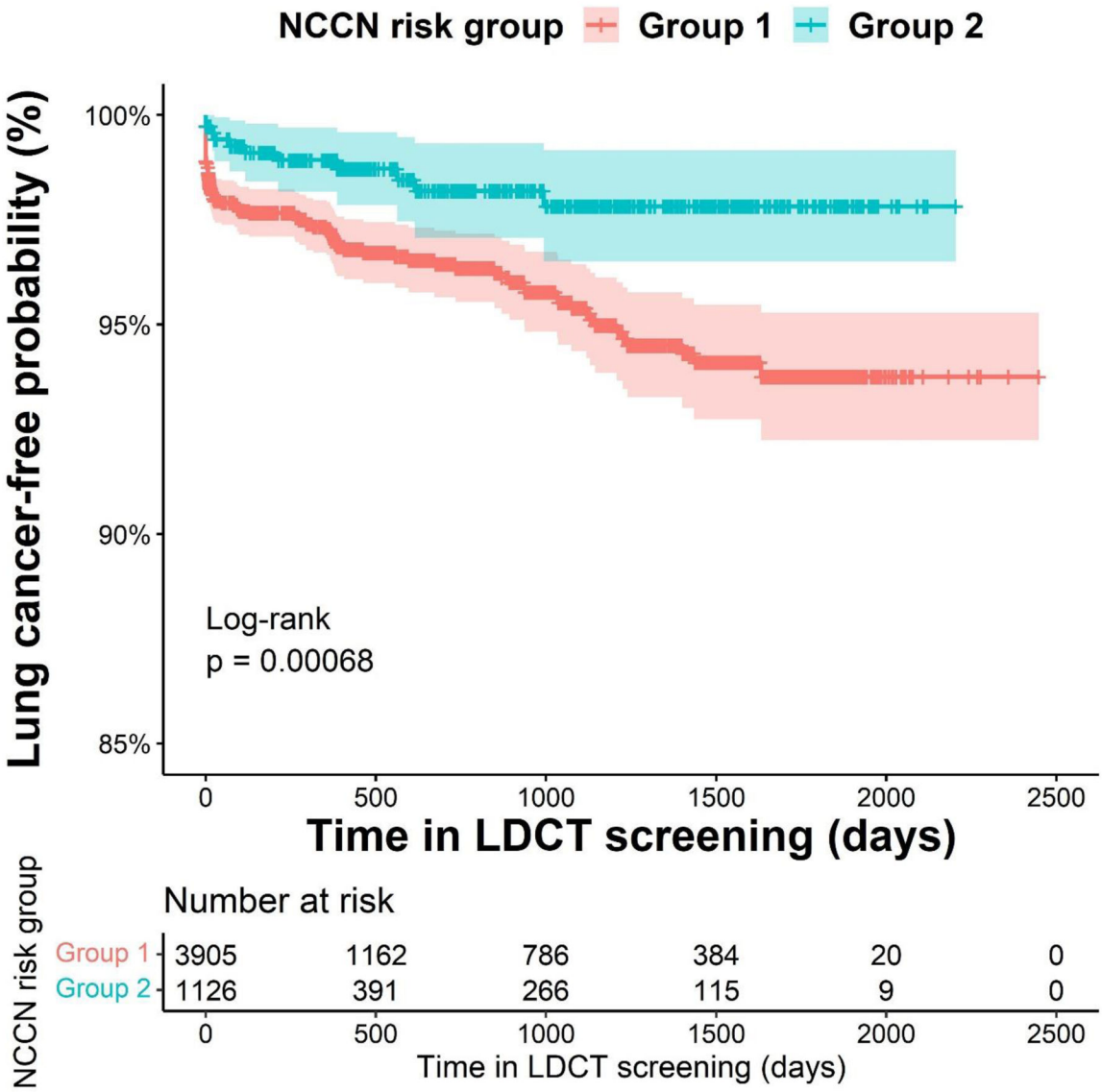


Figure 2.

Kaplan-Meier curve for lung cancer-free probability among 5031 participants enrolled in Highlands Oncology's low-dose computed tomography (LDCT) screening program from 2013 – 2020, stratified according to the National Comprehensive Cancer Network's lung cancer high-risk group definitions.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

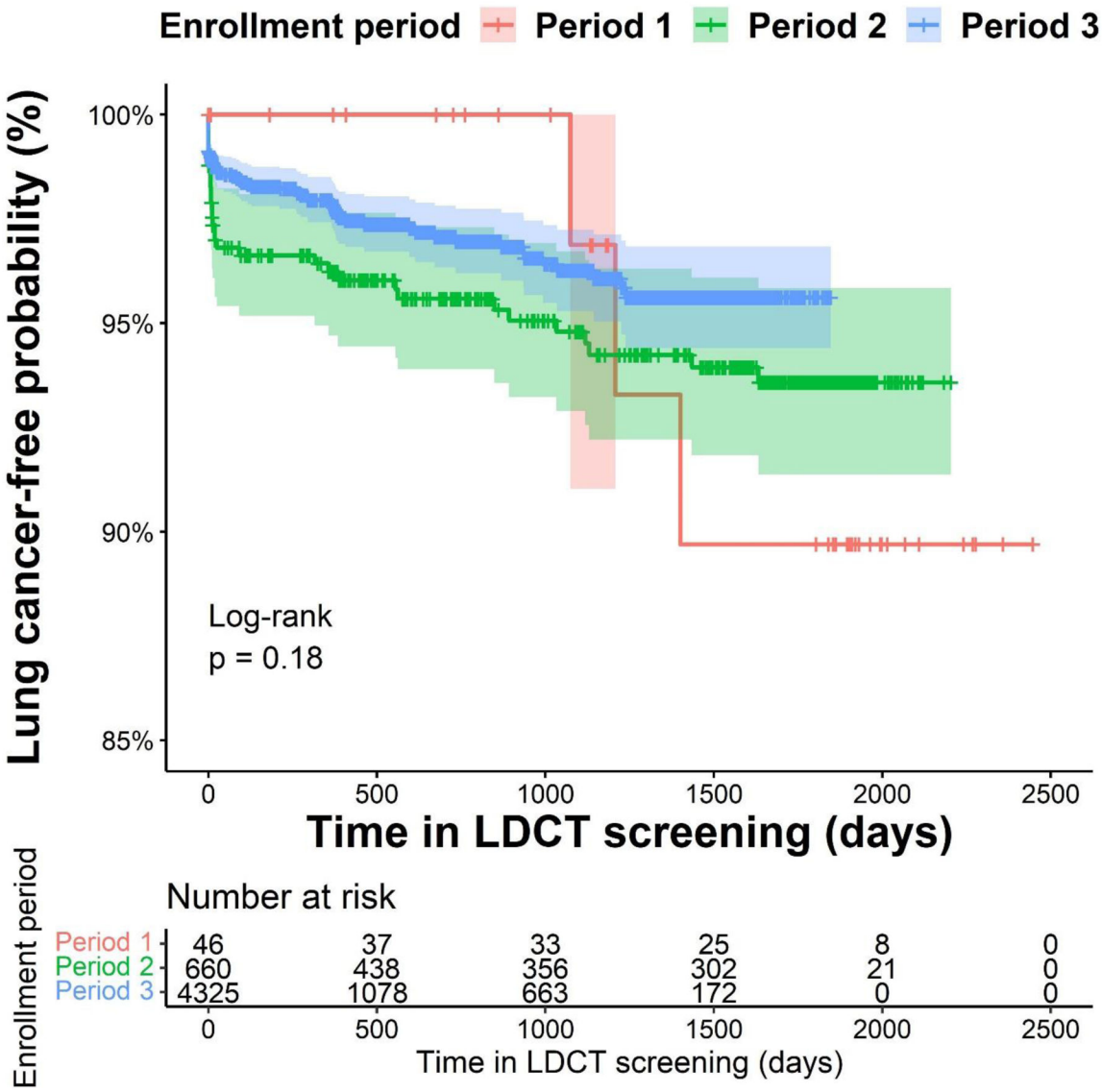


Figure 3.

Kaplan-Meier curve for lung cancer-free probability among 5031 participants enrolled in Highlands Oncology's low-dose computed tomography screening program from 2013 – 2020, stratified according to period of initial enrollment.

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 1.

Characteristics of patients in Highlands Oncology's low-dose computed tomography screening program by period of enrollment from 2013 – 2020.

Characteristic	Period 1 (n = 47)	Period 2 (n = 660)	Period 3 (n = 4328)	p-value
Sex, No. (%)				1.00
Male	22 (47.8)	319 (48.3)	2084 (48.2)	
Female	24 (52.2)	341 (51.7)	2241 (51.8)	
Risk Group, No. (%)				0.39
Category 1	37 (80.4)	525 (79.6)	3343 (77.3)	
Category 2	9 (19.6)	135 (20.4)	982 (22.7)	
Smoking Status, No. (%)				0.01 *
Current Smoker	31 (66.0)	333 (50.5)	2420 (55.9)	
Former Smoker	16 (34.0)	327 (49.5)	1908 (44.1)	
Pack-Years Smoked[a]				< 0.001 †
Median (IQR)	40.0 (25.0)	46.0 (24.0)	51.0 (39.5)	
Age, Years				0.01 †
Mean (SD)	61.7 (6.4)	64.2 (6.9)	63.5 (7.0)	
Range	51 – 79	50 – 81	50 – 92	

No. = Number

IQR = Interquartile Range

SD = Standard Deviation

[a] Log-transformed for ANOVA F-test calculation only

* Indicates statistical significance ($P < 0.05$) using chi-square test for homogeneity

† Indicates statistical significance ($P < 0.05$) using ANOVA F-test

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 2.

Lung cancer and other cancer diagnoses,[a] average enrollment per month, and incidence densities of patients enrolled in Highlands Oncology's low-dose computed tomography screening program by enrollment period from 2013 – 2020.

	Period 1 (n = 46)	**Period 2 (n = 660)**	**Period 3 (n = 4325)**
Stage I, No. (%)	2 (4)	22 (3)	50 (1)
Stage II-IV, No. (%)	1 (2)	10 (2)	33 (1)
Other Cancers, No. (%)	0	8 (1)	25 (1)
Average Enrollment per Month	4.6	66.0	69.8
Total Patient-Years	174.11	1925.21	4231.65
Incidence Density (Per 1,000 Person-Years)	17.2	20.8	25.5

No. = Number

[a]Other cancers include mediastinal or breast cancer diagnosis

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 3.

Multivariable Cox proportional hazard ratios for all lung cancer diagnoses and cause-specific hazards of Stage I or Stage II-IV diagnoses as the result of Highlands Oncology's low-dose computed tomography screening from 2013 – 2020 adjusted for enrollment period, risk group, sex, smoking status, age, and natural log-transformed pack-years.

	Stage I-IV Diagnosis		Stage I Diagnosis		Stage II-IV Diagnosis	
Variable	**HR (95% CI)**	**p-value**	**HR (95% CI)**	**p-value**	**HR (95% CI)**	**p-value**
Enrollment Period						
Period 1	1.88 (0.59 to 6.03)	0.29	2.00 (0.50 to 7.92)	0.32	1.70 (0.26 to 11.02)	0.58
Period 2	1.41 (0.92 to 2.16)	0.11	1.57 (0.92 to 2.69)	0.10	1.16 (0.56 to 2.44)	0.69
Period 3	Ref.	--	Ref.	--	Ref.	--
Risk Group						
Category 1	2.26 (1.20 to 4.26)	0.01	1.72 (0.80 to 3.72)	0.17	4.06 (1.33 to 12.35)	0.01
Category 2	Ref.	--	Ref.	--	Ref.	--
Sex						
Male	0.93 (0.64 to 1.35)	0.71	0.65 (0.40 to 1.04)	0.07	1.71 (0.93 to 3.14)	0.08
Female	Ref.	--	Ref.	--	Ref.	--
Smoking Status						
Current Smoker	1.69 (1.14 to 2.51)	0.009	1.81 (1.08 to 3.03)	0.02	1.50 (0.81 to 2.76)	0.19
Former Smoker	Ref.	--	Ref.	--	Ref.	--
Age, Per Year	1.07 (1.04 to 1.10)	<0.001	1.06 (1.02 to 1.10)	0.001	1.08 (1.02 to 1.14)	0.007
Pack-Years, log-transformed	1.16 (0.76 to 1.78)	0.49	1.23 (0.76 to 1.98)	0.40	1.06 (0.62 to 1.83)	0.83

HR = Hazard Ratio

CI = Confidence Interval

Ref. = Reference group